ITA301 — Information System Analysis & Design
Summer 2026 | Group 3

D2 — SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

AgriUAV Platform

Precision Agricultural Drone Spraying Coordination Platform for the Mekong Delta

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1. Introduction

1.1 Purpose

This document is the **Software Requirements Specification (SRS)** for the **AgriUAV Platform** — Precision Agricultural Drone Spraying Coordination Platform for the Mekong Delta, developed in accordance with **IEEE Std 830-1998**.

Purpose of this SRS:

- Provide complete and clear requirements specifications for the development team (System Designer, System Analyst) to design and implement the system according to business needs.
- Serve as a signed agreement between the analysis team and stakeholders (Cooperatives, Plant Protection Department) on the scope and features of the system.
- Serve as the basis for building test cases, traceability matrix, and subsequent deliverables (D3 Models, D4 SDD).

1.2 Scope

System Name: AgriUAV Platform

AgriUAV Platform is a Web (Dashboard) and Mobile App platform that helps Agricultural Cooperatives in the Mekong Delta automate the scheduling, coordination, and monitoring of fleets of plant-protection drones with precision.

In-Scope: Manage a drone fleet of up to 50 units per cooperative; spray scheduling; field mapping; automatic assignment; real-time tracking; geofencing; export compliance reports for Decree 288/2025/ND-CP.

Out-of-Scope: Direct drone control (fly-by-wire); AI pest detection; integration with actual drone hardware; financial/invoicing management; UTM flight permit applications.

1.3 Definitions, Acronyms, Abbreviations

See Appendix A — Glossary at the end of this document.

1.4 References

- Nguyen, D.L. et al. (2026). 'The Effect of Farm Size on the Decision to Adopt Digital Technology: UAVs in Rice Production in Vietnam.' Research on World Agricultural Economy.
- Decree 288/2025/ND-CP — Regulation of unmanned aerial vehicle flight operations (effective 05/11/2025).
- Vietnam News (2026). 'Use of agricultural drones on the rise in the Mekong Delta.'
- IEEE Std 830-1998 — IEEE Recommended Practice for Software Requirements Specifications.
- D1 AgriUAV Platform — Project Proposal (ITA301 Summer 2026, Group 3).

1.5 Overview

This SRS document is organized according to IEEE 830-1998 and includes the following sections:

- (2) Overall system description
- (3) Functional (FR) and Non-Functional (NFR) requirements specification
- (4) External interfaces
- (5) Use Cases
- (6) User Stories
- Glossary Appendix.

2. Overall Description

2.1 Product Perspective

AgriUAV Platform fills the gap of the **Platform Layer** in the agricultural UAV ecosystem in the Mekong Delta:

Layer	Current State	AgriUAV Role
Hardware	DJI, XAG, YANMAR — saturated market	Current: drone position is received indirectly via the pilot's smartphone GPS (see Assumption in Section 2.5 and Section 4.2). Future: direct MAVLink telemetry integration with DJI/XAG drones.
Service	Cooperatives, TT Drone — manual operation (Zalo/Excel)	Digitize and automate operations
Platform	**AgriUAV Platform — Market gap currently being filled**	

2.2 Product Functions (Summary)

- Field Mapping: Digitize field maps with GPS, automatically calculate pesticide dosages.
- Scheduling: Book spraying sessions online, check for schedule conflicts, automatic confirmation.
- Fleet Dispatch: Automatically assign drones and pilots based on an optimization algorithm.
- Real-time Monitoring: Track drone positions, geofence alerts, monitor battery and pesticide levels.
- Compliance: Automatically store flight logs, export Plant Protection Department reports per Decree 288/2025.
- Notification: Send SMS/Zalo status updates to farmers.

2.3 User Characteristics

User	IT Skill	Characteristics & UX Requirements
Farmer	Low (basic smartphone)	Average age ~51 (Nguyen 2026). The app needs intuitive icons, large text ≥16pt, ≤5-step booking, offline mode.
HTX Manager	Medium (Excel, Zalo)	Needs an overview Dashboard, calendar view, 1-click reports. Uses Web on desktop.
Operator	Medium (smartphone)	Uses Mobile App in the field. Needs GPS check-in, clear alerts, offline operation.
Agronomist	High (Excel, GIS)	Needs to configure spray formulas, detailed field mapping, view historical data.

2.4 Constraints

- The system only assigns pilots holding valid flight certificates under Decree 288/2025/ND-CP.
- The Mobile App must support basic offline operation (view schedule, check-in/out) when 4G is unavailable in the field.
- Flight log data must be retained for at least 5 years as required by law.
- The system does not directly interfere with drone hardware (no fly-by-wire, no autopilot).
- Maximum scope: 50 drones per cooperative — not designed for nationwide scale in the initial phase.

2.5 Assumptions and Dependencies

- Assumption: Drones have been registered and assigned valid identifiers under Decree 288/2025 before being added to the system.
- Assumption: Pilots already hold flight certificates issued by competent authorities before an Operator account is created.
- Assumption: MAVLink telemetry integration with real drones is OUT OF SCOPE for the initial phase — drone position is updated via the pilot's Mobile App GPS.

- Dependency: The system depends on a third-party Weather API (OpenWeatherMap). If the API is down, weather alert features will not function.
- Dependency: The Zalo notification feature depends on the Zalo Official Account API.

3. Specific Requirements

3.1 Functional Requirements (FR)

The table below lists **20 Functional Requirements** classified by module, with Priority following MoSCoW (Must / Should / Could). Source specifies the originating stakeholder or pain point of the requirement.

ID	Requirement Description	Module	Priority	Source
FR01	The system allows the HTX Manager to create, edit, and store field maps with GPS coordinates, area (ha), and crop type.	Field Mapping	**Must**	HTX Manager, Pain Point Step 1
FR02	The system calculates and displays pesticide dosages for each field zone based on area, pest type, and spray formulas configured by the Agronomist.	Field Mapping	**Must**	Agronomist, Plant Protection Regulations
FR03	The system provides a spraying-booking function for farmers: choose field, spray date/time, pesticide type, and confirm the order in ≤5 steps.	Scheduling	**Must**	Farmer, Pain Point Step 2
FR04	The system automatically checks for schedule conflicts and displays warnings when multiple spraying orders overlap in time on the same area.	Scheduling	**Must**	HTX Manager, Pain Point Step 2
FR05	The HTX Manager can view, edit, and confirm spraying schedules in a calendar view (by day / week / month).	Scheduling	**Should**	HTX Manager
FR06	The system automatically suggests drone assignments for each spraying order based on: drone status (idle / flying / under maintenance), remaining tank capacity, and proximity to the field.	Fleet Dispatch	**Must**	HTX Manager, Pain Point Step 3
FR07	The system manages pilot records including: full name, flight certificate number, certificate expiration date, flight history, and operational status.	Fleet Dispatch	**Must**	Decree 288/2025/ND-CP — FR-REG01
FR08	The Operator/Pilot receives the flight assignment notification via the Mobile App with complete information: field, GPS coordinates, pesticide dosage, and required time.	Fleet Dispatch	**Must**	Operator/Pilot
FR09	The Pilot can perform on-site check-in (GPS-verified) and check-out upon completion; the system automatically records the actual time and location.	Fleet Dispatch	**Must**	HTX Manager, fraud prevention
FR10	The system displays the position and status of all active drones on a real-time map, updated at least every 3 seconds.	Monitoring	**Must**	HTX Manager, Pain Point Step 4

ID	Requirement Description	Module	Priority	Source
FR11	The system automatically issues alerts (push notification + dashboard display) when it detects: (a) drone battery below 20%, (b) pesticide below the completion threshold, (c) drone leaving the geofence.	Monitoring	**Must**	Safety, Pain Point Step 4
FR12	The system sets up and enforces automatic geofencing based on field boundaries, including a 50m buffer zone around marked high-voltage power lines.	Monitoring	**Must**	Safety — Long An 110kV incident
FR13	The system automatically stores complete flight logs including: drone identifier, pilot ID, flight time, GPS coordinates, and volume of pesticide sprayed.	Compliance	**Must**	Decree 288/2025/ND-CP — FR-REG01
FR14	The system exports seasonal reports in the Plant Protection Department format (PDF/Excel) with complete information: spray area, pesticide type, dosage, and confirmation that the pilot holds a valid certificate.	Compliance	**Must**	Plant Protection Department — FR-REG02, Decree 288
FR15	The HTX Manager can retrieve the spray history of each field by time range and export data in VietGAP/GlobalGAP format for export companies.	Compliance	**Should**	Agricultural export companies
FR16	The system manages pesticide inventory: stock entry, stock-level tracking, low-stock alerts (below configured threshold), and verification that pesticides are on the Plant Protection Department's approved list.	Inventory	**Should**	HTX Manager, Agronomist
FR17	The system sends spraying-order status updates to farmers via SMS and/or Zalo: booking confirmation, spraying started, spraying completed.	Notification	**Should**	Farmer
FR18	The system enforces role-based access control with 4 roles: Admin (full access), HTX Manager (manages their own cooperative), Operator (views personal schedule + check-in/out), Farmer (books spraying + views progress).	User Mgmt	**Must**	All stakeholders
FR19	The Mobile App and Web Dashboard support dark mode, which can be toggled manually or synced with the device's system theme.	UX	**Could**	Operator (early-morning / late-evening flights, anti-glare)
FR20	The system supports multiple languages — at minimum Vietnamese (default) and English — with runtime switching that does not require re-login.	i18n	**Could**	Export companies, foreign investors

Must	Mandatory — the system is incomplete without it	**Should**	Important — should be in the MVP	**Could**	Optional — only if time allows

3.2 Non-Functional Requirements (NFR)

The table below specifies **7 Non-Functional Requirements** with specific quantitative thresholds, calibrated to the actual context of the Mekong Delta.

ID	Category	Criterion	Quantitative Threshold	Rationale / Source
NFR01	**Performance**	Drone position update on the map	Update cycle ≤3 seconds for up to 50 concurrent drones. Data transmission latency <500ms under normal 4G conditions.	DJI Agras T40 drone speed: ~5–7 m/s; 3 seconds is sufficient for safety alerts
NFR02	**Reliability**	System availability during peak season	Uptime ≥99% (allowing max downtime ~7.2 hours/year). RTO (Recovery Time Objective) ≤15 minutes after an incident.	Peak rice season: 30–60 days, extended downtime is unacceptable
NFR03	**Usability**	Farmer completes a spraying booking	First-time users complete the booking task in ≤5 minutes with no assistance. The Mobile App interface supports font ≥16pt and intuitive icons for users with limited IT experience (average age ~51).	Nguyen et al. 2026: Mekong Delta farmers' average age is 51, with limited IT experience
NFR04	**Scalability**	Drone fleet size and concurrent users	Supports up to 50 concurrent drones per cooperative. The system handles up to 500 requests per second at peak (season). Stores flight log data for at least 5 years.	Scope: max 50 drones per cooperative; 500 req/s suffices for 10 cooperatives concurrently
NFR05	**Security**	Data security and access control	Data-in-transit encryption: TLS 1.3. Sensitive data-at-rest encryption: AES-256. Authentication: JWT + Refresh Token. RBAC (4 roles). Complete audit-trail logging.	Decree 288/2025 on airspace data security; OWASP Top 10
NFR06	**Availability**	Basic offline operation in the field when 4G is unavailable	The Mobile App maintains offline mode: view cached flight schedule, offline check-in/out (synced when connection is restored). Automatic sync within ≤30 seconds when connectivity returns.	Mekong Delta reality: remote areas often lack 4G; pilots need to check in directly in the field
NFR07	**Maintainability**	Maintainability and system update capability	Hotfix deployment time ≤4 hours. Loosely coupled modules so that updating one module does not affect others. Test code coverage ≥70%.	Decree 288 may change → the Compliance module needs to be updated quickly

4. External Interface Requirements

4.1 User Interfaces

- Web Dashboard (HTX Manager, Admin): React.js / Vue.js, supports the latest Chrome, Firefox, and Safari browsers. Responsive from 1280px. Map integration via Leaflet.js.
- Mobile App (Farmer, Operator): React Native / Flutter, supports Android ≥8.0 and iOS ≥14. Offline-first architecture.
- Accessibility requirements: Font ≥16pt on mobile, contrast ratio ≥4.5:1, icons with label text.

4.2 Hardware Interfaces

- Smartphone GPS module: accuracy ≤5m for check-in verification.
- Smartphone camera: capture on-site photos attached to completion reports (optional).
- MAVLink telemetry interface: out of MVP scope — recorded in assumptions for a future version.

4.3 Software Interfaces

Interface	Description & Technical Specification
GPS / Location Service	The Mobile App uses native GPS (iOS CoreLocation / Android LocationManager). Required accuracy: ≤5m. Used to verify pilot check-ins and display drone positions.
Weather API	Weather REST API integration (OpenWeatherMap or equivalent). Polling every 15 minutes. Response: wind speed, rainfall, 2-hour forecast. Format: JSON.
SMS / Zalo Notification	Send SMS via Twilio or VNPT SMS Gateway. Send Zalo OA messages via the Zalo Official Account API. Triggers: successful booking, spraying started, spraying completed.
MAVLink Telemetry (future)	MAVLink v2 protocol to receive telemetry from DJI/XAG drones if hardware is integrated (outside current D1 scope — recorded in assumptions). Data: GPS, altitude, battery, speed.
Map / GIS	Leaflet.js or Google Maps SDK to display field maps and drones. Supports drawing polygons (field boundaries) and displaying drone GPS trails.
Export Format	Exported reports: PDF (pdfmake/iTextSharp) and Excel (Apache POI/ExcelJS). Template follows the Plant Protection Department's format. VietGAP/GlobalGAP: standard JSON/CSV format for data sharing.

4.4 Communication Interfaces

- Backend ↔ Frontend: REST API over HTTPS (TLS 1.3). JSON format.
- Real-time updates: WebSocket (Socket.IO) for drone position tracking (≤3s update cycle).
- Push notifications: Firebase Cloud Messaging (FCM) for Android/iOS.
- SMS: HTTPS POST to Twilio API / VNPT SMS Gateway.

5. Use Cases

This section describes **10 main Use Cases** of the AgriUAV Platform, including the main flow and ≥2 alternative flows for each critical UC.

UC01 — Book a Spraying Session

Use Case ID	**UC01**
Use Case Name	Book a Spraying Session
Primary Actor	Farmer
Secondary Actor	HTX Manager (confirmation)
Preconditions	1. Farmer is logged in. 2. The Farmer's field has been created in the system. 3. At least 1 drone in the fleet is available.
Postconditions (Success)	The spraying order is saved with status 'Pending confirmation'. The HTX Manager receives a notification to assign a drone.
Postconditions (Failure)	If the Farmer does not re-confirm the order within 24 hours of creation, the system automatically cancels the order, changes its status to 'Cancelled — Confirmation timeout', notifies the Farmer, and releases the reserved time slot. If the system cannot create the order due to a technical fault (DB error, pesticide-catalog API unavailable), the status remains unchanged and the Farmer receives an error notification.
Main Flow	1. The Farmer selects a field from the list. 2. The Farmer selects the desired spray date and time. 3. The system checks availability and displays free time slots. 4. The Farmer selects a pesticide from the approved list. 5. The Farmer confirms the booking; the system creates the order and sends SMS/Zalo notifications.
Alternative Flow	Alt 3a — Schedule conflict: The system warns about the conflict and suggests the 3 nearest available time slots. Alt 4a — Pesticide not on the approved list: The system displays a red warning and does not allow order confirmation.
Business Rule	Bookings must be made at least 24 hours before the spraying time. No bookings when forecast wind speed > 3 m/s.
Priority	Must — Linked: FR03, FR04, FR17

UC02 — Automatic Drone Assignment for a Spraying Order

Use Case ID	**UC02**
Use Case Name	Automatic Drone Assignment for a Spraying Order
Primary Actor	HTX Manager
Secondary Actor	System (automatic suggestion)
Preconditions	1. A spraying order has been created (UC01) and is in 'Pending confirmation' status. 2. At least 1 drone and 1 valid pilot (with a non-expired certificate) are available.
Postconditions (Success)	A drone and a pilot are assigned. The Pilot receives a notification on the Mobile App. The order moves to 'Assigned'.
Main Flow	1. The HTX Manager opens the list of spraying orders pending assignment. 2. The HTX Manager selects an order to process.

	3. The system automatically suggests the most suitable drone (based on: nearest location, sufficient battery, remaining tank). 4. The system automatically suggests a suitable pilot (valid certificate, free schedule). 5. The HTX Manager confirms or modifies the suggestion. 6. The system updates the order status to 'Assigned' and notifies the Pilot.
Alternative Flow	Alt 3a — No drone available: The system notifies the user and suggests rescheduling to a different time slot. Alt 4a — Pilot lacks a valid certificate: The system hides that pilot's name and displays the reason.
Business Rule	Pilots must hold a valid flight certificate under Decree 288/2025. Drones under maintenance are not assigned.
Priority	Must — Linked: FR06, FR07, FR08, FR13

UC03 — On-site Check-in (GPS-verified)

Use Case ID	**UC03**
Use Case Name	On-site Check-in (GPS-verified)
Primary Actor	Operator / Pilot
Secondary Actor	—
Preconditions	1. The Pilot has received the flight assignment (UC02) and is logged into the Mobile App. 2. The Pilot is on-site at the field (within 100m of the field's GPS coordinates). 3. The drone has been prepared and is ready to fly.
Postconditions (Success)	Check-in is recorded with timestamp and GPS. The HTX Manager sees the real-time update on the dashboard. The geofence is automatically activated.
Main Flow	1. The Pilot opens the Mobile App and selects today's spraying order. 2. The Pilot taps the 'On-site Check-in' button. 3. The system verifies the Pilot's GPS position (within 100m of the field's coordinates). 4. The Pilot enters pre-flight parameters: pesticide level in the tank, drone condition. 5. The system records the check-in time and GPS position, and updates the order status to 'In progress'.
Alternative Flow	Alt 3a — Pilot's GPS is outside the 100m radius: The system warns and requires manual confirmation with a stated reason. Alt 4a — Offline (no 4G): The system stores the check-in locally and syncs once connectivity returns (≤30 seconds).
Business Rule	GPS verification radius: 100m. Check-in is only allowed within ±30 minutes of the scheduled spraying time.
Priority	Must — Linked: FR09, FR10, FR12

UC04 — Real-time Spraying Progress Monitoring

Use Case ID	**UC04**
Use Case Name	Real-time Spraying Progress Monitoring
Primary Actor	HTX Manager
Secondary Actor	System (auto-update)
Preconditions	1. At least 1 spraying order is in 'In progress' status (checked in). 2. The HTX Manager is logged into the Web Dashboard.

Postconditions (Success)	The HTX Manager has full real-time information on the progress of the entire drone fleet.
Main Flow	1. The HTX Manager opens the 'Monitoring Map' Dashboard. 2. The system displays the positions of all flying drones on the map (updated ≤3 seconds). 3. The HTX Manager selects a drone to view details: % area completed, remaining pesticide, estimated completion time. 4. The HTX Manager can view the drone's flight trail. 5. The system displays a red alert if a drone violates the geofence.
Alternative Flow	Alt 2a — Drone connection lost (>15 seconds without signal): The system displays a 'Disconnected' status and alerts the HTX Manager. Alt 5a — Drone leaves the geofence: The system immediately triggers an audio alert + popup on the dashboard and sends a push notification to the Pilot.
Business Rule	Position data is logged with a timestamp every 3 seconds to support later legal reporting.
Priority	Must — Linked: FR10, FR11, FR12, FR13

UC05 — Receive and Handle Low-Pesticide / Low-Battery Alerts

Use Case ID	**UC05**
Use Case Name	Receive and Handle Low-Pesticide / Low-Battery Alerts
Primary Actor	Operator / Pilot
Secondary Actor	HTX Manager
Preconditions	1. The Pilot is performing spraying (checked in — UC03). 2. Alert thresholds are configured (battery: 20%, pesticide: enough to finish the current zone).
Postconditions (Success)	The alert is logged with a timestamp. Order status is updated accordingly (Paused / In progress / Completed early).
Main Flow	1. The system detects drone battery dropping below 20% (or pesticide insufficient for the next zone). 2. The system sends a push notification to the Pilot and an alert on the HTX Manager's dashboard. 3. The Pilot confirms receipt of the alert. 4. The Pilot lands, refills pesticide / replaces battery, then taps 'Resume spraying'. 5. The system records the pause time and recalculates the completion percentage.
Alternative Flow	Alt 3a — The Pilot does not respond within 5 minutes: The system escalates the alert to the HTX Manager and turns the drone icon red. Alt 4a — The Pilot decides to terminate spraying entirely: The Pilot taps 'End early', and the system records the reason and the area already sprayed.
Business Rule	Battery and pesticide alert thresholds are configured by the HTX Admin; defaults: battery 20%, pesticide 15%.
Priority	Must — Linked: FR10, FR11

UC06 — Adverse Weather Alert and Handling of In-flight Orders

Use Case ID	**UC06**
Use Case Name	Adverse Weather Alert and Handling of In-flight Orders
Primary Actor	System (automatic)
Secondary Actor	HTX Manager, Pilot

Preconditions	1. At least 1 spraying order is in progress or scheduled to start (within the next 2 hours). 2. The system is integrated with weather data (API).
Postconditions (Success)	Decisions and their rationale are fully logged. The Pilot receives the final decision notification.
Postconditions (Failure)	If the Weather API does not respond for more than 30 minutes (see Dependency in Section 2.5), the system switches to a 'Weather alert — Unavailable' state and displays a banner for the HTX Manager to make a manual decision; orders are NOT automatically suspended. If neither the HTX Manager nor the Pilot responds to an alert within 30 minutes when wind ≥3 m/s, the system forces the order status to 'Suspended — Lacking safety confirmation' to safeguard legal compliance.
Main Flow	1. The system pulls weather data every 15 minutes from the API. 2. The system detects wind speed ≥3 m/s or rain forecast within the next 30 minutes. 3. The system automatically sends alerts to the Pilot (in flight) and the HTX Manager. 4. The HTX Manager reviews and decides: halt spraying or continue with risk acknowledgement. 5. If halted: the system updates the order status to 'Suspended — Weather' and suggests an alternative time slot.
Alternative Flow	Alt 4a — The HTX Manager does not respond within 10 minutes: The system automatically marks the order as 'Needs review' and continues to send reminders every 5 minutes. Alt 4b — The HTX Manager confirms continuation: The system records the decision + timestamp + confirmer's name in the audit trail.
Business Rule	Wind threshold: ≥3 m/s (per DJI Agras agricultural-drone spraying standards). Weather data from the OpenWeatherMap API or equivalent.
Priority	Should — Linked: FR11, FR17

UC07 — Check-out and Spraying Completion Record

Use Case ID	UC07
Use Case Name	Check-out and Spraying Completion Record
Primary Actor	Operator / Pilot
Secondary Actor	System
Preconditions	1. The Pilot has checked in (UC03) and finished spraying on-site. 2. The Pilot has network connectivity (or is offline — will sync later).
Postconditions (Success)	A complete flight log is saved to the database with full legal information. The Farmer receives a completion notification. Pesticide inventory is updated.
Main Flow	1. The Pilot taps 'Complete spraying' on the Mobile App. 2. The Pilot enters actual data: pesticide used, actual area sprayed, abnormal observations (if any). 3. The system calculates and displays a summary: sprayed area, pesticide used, flight duration, comparison with the plan. 4. The Pilot confirms and signs digitally (fingerprint/PIN). 5. The system automatically generates the flight log and updates the order status to 'Completed'. 6. The system sends a notification to the Farmer via SMS/Zalo.
Alternative Flow	Alt 2a — Actual pesticide usage deviates >20% from the plan: The system requires the Pilot to enter a reason before confirmation. Alt 4a — Offline: The system stores the check-out locally and syncs within ≤30 seconds once connectivity returns.

Business Rule	The flight log includes: drone identifier, pilot ID, GPS trail, pesticide volume — mandatory under Decree 288/2025.
Priority	Must — Linked: FR09, FR13, FR16, FR17

UC08 — Export Plant Protection Department Compliance Report

Use Case ID	UC08
Use Case Name	Export Plant Protection Department Compliance Report
Primary Actor	HTX Manager
Secondary Actor	Plant Protection Department (report recipient)
Preconditions	1. The HTX Manager is logged in. 2. Spraying log data exists for the reporting period (monthly/quarterly/seasonal).
Postconditions (Success)	The PDF/Excel report is downloaded. The export history is saved to the system's audit log.
Main Flow	1. The HTX Manager selects 'Export compliance report' from the Compliance menu. 2. The HTX Manager selects the time range and target cooperative. 3. The system aggregates data: list of drones (with identifiers), list of pilots (with certificate numbers), total spray area per pesticide type. 4. The system generates the PDF and Excel report files simultaneously. 5. The HTX Manager previews and downloads the report. The system saves the export history (who, when).
Alternative Flow	Alt 3a — A pilot flew but their certificate expired during the period: The system highlights the violating flights in red in the report and creates a 'Compliance note' section. Alt 4a — Insufficient data (missing logs): The system lists the spraying orders with missing data and refuses to export the report until they are complete.
Business Rule	The report format follows the Plant Protection Department template (FR-REG02). The report must be digitally signed or confirmed by the HTX Manager.
Priority	Must — Linked: FR13, FR14, FR15

UC09 — Field Map Management (Field Mapping)

Use Case ID	UC09
Use Case Name	Field Map Management (Field Mapping)
Primary Actor	HTX Manager / Agronomist
Secondary Actor	—
Preconditions	1. The user is logged in with the role HTX Manager or Agronomist. 2. The Farmer is registered in the system.
Postconditions (Success)	The field is saved with complete coordinates, area, and spray formula. The Farmer can book spraying for this field.
Postconditions (Failure)	If the polygon is not closed or the area is outside the limits (<0.1 ha or >100 ha), the system does not save the field and displays a specific error; the Farmer's account is not associated with the erroneous field. If an overlap with an existing field is detected (Alt 3a) and the user does not accept an adjustment, the operation is cancelled and the map is rolled back to its previous state (safe rollback).
Main Flow	1. The HTX Manager selects 'Add new field'. 2. The user draws the field boundary on the map (or enters GPS coordinates).

	3. The system automatically computes the area (ha) from the coordinates. 4. The user enters information: field name, owner (Farmer), crop type, seasonal schedule. 5. The Agronomist configures the spray formula: pesticide type, dosage (L/ha), number of sprays per season. 6. The system saves the map and associates it with the Farmer's account.
Alternative Flow	Alt 2a — Invalid manual GPS coordinates: The system reports a format error and requests re-entry. Alt 3a — The new field overlaps an existing one: The system warns and displays the overlapping region on the map.
Business Rule	Field boundaries must be closed polygons. Minimum area: 0.1 ha. Maximum: 100 ha per field.
Priority	Must — Linked: FR01, FR02

UC10 — View Spraying History and Export VietGAP Data

Use Case ID	UC10
Use Case Name	View Spraying History and Export VietGAP Data
Primary Actor	Farmer / Export Company
Secondary Actor	—
Preconditions	1. The user is logged in. 2. The field has at least 1 completed spraying record.
Postconditions (Success)	Spraying history data is displayed/exported successfully. Access is logged.
Main Flow	1. The user selects the field whose history they want to view. 2. The system displays the spraying sessions: date, pesticide type, dosage, pilot, drone, area. 3. The user filters by time range or pesticide type. 4. The export company selects 'Export VietGAP/GlobalGAP data'. 5. The system generates an Excel/PDF file with full traceability information.
Alternative Flow	Alt 4a — The export company has not been granted access: The system requires the HTX Manager to approve sharing permission first.
Business Rule	Data is only shared after HTX Manager approval. Exports are recorded in the audit log: who, when, what data.
Priority	Should — Linked: FR14, FR15

5.1 Traceability Matrix FR ↔ UC

The table below is the reverse of the 'Linked FR' column shown in each UC — each FR is traced to the UC(s) that use it, demonstrating that no FR is 'orphaned'. This table supports consistency checking and is a ready reference for the Oral Vivas.

FR ID	Short Description	Priority	UCs Using This FR
FR01	Create/edit field maps (Field Mapping)	**Must**	UC09
FR02	Calculate per-zone pesticide dosage	**Must**	UC09
FR03	Book spraying in ≤5 steps (Farmer)	**Must**	UC01
FR04	Schedule-conflict alerts	**Must**	UC01
FR05	Calendar view (HTX Manager)	**Should**	UC01, UC02 (indirectly)

FR ID	Short Description	Priority	UCs Using This FR
FR06	Automatic drone-assignment suggestions	**Must**	UC02
FR07	Pilot records (certificate, expiration)	**Must**	UC02, UC08
FR08	Assignment notifications to the Operator	**Must**	UC02
FR09	GPS-verified Check-in/Check-out	**Must**	UC03, UC07
FR10	Real-time map ≤3s	**Must**	UC03, UC04, UC05
FR11	Battery / pesticide / geofence alerts	**Must**	UC04, UC05, UC06
FR12	Geofencing (50m buffer around power lines)	**Must**	UC03, UC04
FR13	Complete flight logs (Decree 288)	**Must**	UC02, UC04, UC07, UC08
FR14	Export Plant Protection Department reports	**Must**	UC08, UC10
FR15	Export VietGAP/GlobalGAP	**Should**	UC08, UC10
FR16	Manage pesticide inventory	**Should**	UC07
FR17	SMS/Zalo notifications to Farmers	**Should**	UC01, UC06, UC07
FR18	RBAC permissions (4 roles)	**Must**	All UCs (cross-cutting authorization)
FR19	Dark mode	**Could**	Cross-cutting UX (not tied to a specific UC)
FR20	Multi-language VI/EN	**Could**	Cross-cutting i18n (not tied to a specific UC)

Note: FR19 and FR20 are cross-cutting concerns (UX/i18n) with priority Could and are therefore not tied to a specific UC. FR18 (RBAC) is cross-cutting authorization, used implicitly in every UC via the precondition 'is logged in'. 18 out of 20 FRs (90%) are traced directly to at least 1 UC.

6. User Stories

The User Stories below are written following the **INVEST** standard with acceptance criteria in the **Given-When-Then** format. Priority follows MoSCoW; Size is relatively estimated (S/M/L).

Story ID	US01
User Story	As a Farmer, I want to book a spraying session in ≤5 taps so that I can schedule without calling the HTX office.
Acceptance (Given)	The Farmer is logged into the Mobile App and has a field created in the system.
Acceptance (When)	The Farmer selects a field → selects date/time → selects a pesticide → taps 'Confirm'.
Acceptance (Then)	The system displays a booking confirmation and sends an SMS confirmation to the Farmer within <30 seconds.
Priority / Size	Must | S (Small)
Linked FR	FR03, FR04, FR17

Story ID	US02
User Story	As an HTX Manager, I want the system to auto-suggest the best drone and pilot for each job so that I don't have to manually match resources.
Acceptance (Given)	There is a spraying order pending assignment, and ≥1 drone and pilot are available with valid certificates.
Acceptance (When)	The HTX Manager opens the assignment screen and views the pending order.
Acceptance (Then)	The system suggests the most suitable drone and pilot within <3 seconds. The HTX Manager only needs to tap 'Confirm' or override the suggestion.
Priority / Size	Must | M (Medium)
Linked FR	FR06, FR07, FR08

Story ID	US03
User Story	As an Operator/Pilot, I want to check-in on-site with GPS verification so that my flight log is automatically recorded without paperwork.
Acceptance (Given)	The Pilot is near the field (within 100m) and has a flight scheduled today in the Mobile App.
Acceptance (When)	The Pilot taps 'On-site Check-in' in the Mobile App.
Acceptance (Then)	The system verifies GPS within <5 seconds, records the check-in with timestamp and location, and updates the order status to 'In progress'.
Priority / Size	Must | S (Small)
Linked FR	FR09, FR10, FR13

Story ID	US04
User Story	As an HTX Manager, I want to see all drones on a live map updated every 3 seconds so that I can monitor the fleet from the office.
Acceptance (Given)	There is ≥1 drone in 'In progress' status currently flying.
Acceptance (When)	The HTX Manager opens the 'Monitoring Map' Dashboard.

Acceptance (Then)	All drones are shown on the map at their accurate positions, updated ≤3 seconds. Clicking a drone reveals % completion and remaining pesticide.
Priority / Size	Must | M (Medium)
Linked FR	FR10, FR11, FR12

Story ID	**US05**
User Story	As an HTX Manager, I want to export a compliance report for the Plant Protection Department in PDF/Excel format so that I can meet the reporting deadline under Decree 288/2025.
Acceptance (Given)	The HTX Manager is logged in and has complete spraying-log data for the reporting period.
Acceptance (When)	The HTX Manager selects the reporting period and taps 'Export compliance report'.
Acceptance (Then)	The system generates PDF and Excel files within <30 seconds containing complete information: drone identifier, pilot certificate number, sprayed area, pesticide type. The file follows the Plant Protection Department's format.
Priority / Size	Must | L (Large)
Linked FR	FR13, FR14

Story ID	**US06**
User Story	As an Operator, I want the Mobile App to work offline so that I can check-in and check-out even when there's no 4G signal at the field.
Acceptance (Given)	The Pilot is in an area without 4G but has a pre-synced flight schedule.
Acceptance (When)	The Pilot performs check-in or check-out while offline.
Acceptance (Then)	The app saves the action locally and displays 'Will sync when online'. Once 4G is restored, it syncs automatically within ≤30 seconds.
Priority / Size	Must | M (Medium)
Linked FR	FR09, NFR06

Story ID	**US07**
User Story	As an Agronomist, I want to configure spray formulas per field zone so that the correct pesticide dosage is automatically calculated for each spraying job.
Acceptance (Given)	The Agronomist is logged in, and the field already has a map.
Acceptance (When)	The Agronomist selects a field and opens the 'Configure spray formula' tab.
Acceptance (Then)	The Agronomist enters pesticide type, dosage (L/ha), and number of sprays. The system automatically computes the total pesticide needed for the whole field and shows a preview.
Priority / Size	Should | M (Medium)
Linked FR	FR01, FR02

Story ID	**US08**
User Story	As a Farmer, I want to receive SMS/Zalo updates when my field has been sprayed so that I don't have to call to confirm.

Acceptance (Given)	The Farmer's spraying order has been completed (UC07).
Acceptance (When)	The Pilot checks out and confirms completion.
Acceptance (Then)	The Farmer receives an SMS/Zalo message within <1 minute with the content: 'Field [name] has been sprayed at [time]. Pesticide used: [X] liters.'
Priority / Size	Should | S (Small)
Linked FR	FR17

Appendix A — Glossary (UAV Domain Terminology)

Term / Abbreviation	Definition
AgriUAV Platform	System name — a web and mobile platform that coordinates precision agricultural drone spraying in the Mekong Delta.
Mekong Delta (ĐBSCL)	Đồng bằng sông Cửu Long — Vietnam's largest specialized rice-growing region.
Drone / UAV	Unmanned Aerial Vehicle — in this document refers to agricultural spraying drones (DJI Agras, XAG, etc.).
HTX (Cooperative)	Agricultural cooperative — an organization that manages and provides drone spraying services to farmers.
Operator / Pilot	The person who operates the drone directly in the field, holding a valid flight certificate under Decree 288/2025.
Geofencing	A virtual geographic fence — a boundary defining the safe flight area, triggering an alert when a drone crosses it.
Field Mapping	Digitization of field maps with GPS coordinates and accurate area.
Check-in / Check-out	The Pilot confirms on-site presence (check-in) and confirms spraying completion (check-out) via the GPS-verified Mobile App.
Decree 288/2025/ND-CP	Decree 288/2025/ND-CP — regulating the operation of unmanned aerial vehicles, effective from 05/11/2025.
Plant Protection Department	Plant Protection Department — the state agency in charge of pesticides and crop-pest control.
Operations Department (General Staff)	A unit under the General Staff that issues flight permits and manages drone identifiers under Decree 288/2025.
MAVLink	Micro Air Vehicle Link — a common drone communication protocol, used to receive telemetry (position, battery, speed).
RTK GPS	Real-Time Kinematic GPS — high-accuracy GPS (≤2cm), commonly integrated in high-end agricultural drones.
VietGAP / GlobalGAP	Vietnamese / international Good Agricultural Practice standards — require traceability of pesticide usage.
SaaS	Software-as-a-Service — a software subscription business model billed monthly/yearly (500,000 VND/drone/month).
RBAC	Role-Based Access Control — permissions assigned by role: Admin, HTX Manager, Operator, Farmer.
STRIDE	Security analysis model: Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, Elevation of Privilege.
TLS 1.3	Transport Layer Security v1.3 — the latest standard internet-connection encryption protocol.
BVLOS	Beyond Visual Line of Sight — flight beyond direct line of sight (outside the scope of this system).
SOM / SAM / TAM	Serviceable Obtainable / Addressable Market / Total Addressable Market — market-sizing segmentation tiers.